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Question #1

1- To read 8-bit data from input port with address 3FAH the following instructions can be used:

A. MOV DX,3FAH; IN AL,DX
B. MOV DX,3FAH; OUT AL,DX
C. MOV BX,3FAH; IN AL,BX
D. MOV BX,3FA; OUT AL,BX

2- The interrupt vector contains two words (pointers) which specify the address of:

A. Interrupt vector table
B. Interrupt service routine
C. Interrupt type
D. I/O ports

3- The binary value of AL and carry flag (CF)after the following instructions have executed is

```
MOV AL,10101011B
SAR AL,1
```

A. AL=11010101B CF=1
B. AL=01010101B CF=1
B. AL=01010100B CF=0
D. AL=01010100B CF=0

4- What will be the final value of AX after execution the following codes?

```
     MOV AX,6
     MOV CX, 4
L1:  INC AX
     LOOP L1
```

A. 11
B. 10
C. 9
D. 12

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5- What is the value of AX register after executing the following instructions?

```
MOV AX,FA45H
TEST AL, 1
JNZ L
ADD AL,1
JMP L2
L: XCHG AL,AH
L2: HLT
```

A. 01FAH
B. 45FAH
C. 00FBH
D. 00FAH

6- If there is a carry from lowest 4 bit during addition, ______ flag sets.

A. Carry
B. Auxiliary carry
C. Over flow
D. Sign

7- Find the content of register AL after execution of the following code

```
MOV AL,4
MOV BL,-4
CMP AL,BL
JG L
ADD AL,BL
L: XOR AL, BL
```

A. FFH
B. F8H
C. FBH
D. FDH

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8- What is the content of register Al and CL after execution of the following code?

```
MOV CL,3
MOV AL, 04H
L: ADD AL,CL
INC CL
CMP AL,8
JB L
```

A. Al= 7;CL=4
B. AL= 11;CL=5
C. AL= 11;CL=4
D. AL= 16;CL=5

9- Assume that X memory location defined as below find the content of register DX and BX

```
X DD 13457ABCH
MOV DX,X
MOV BX,235FH
XCHG DL,BH
```

A. DX= 7A23H; BX=BC5FH
B. DX= 1323H;BX=455FH
C. DX= 1323H; BX=7A5FH
D. DX= 7A13H;BX=AB5FH

10- Find the content of memory location X after execution of the following code

```
X DB ?
MOV AX,9
MOV BL,2
DIV BL
OR AH,30H
MOV X,AH
```

A. X= '0'
B. X=0
C. X='1'
D. X=1

11- Assume that the offset address of X is 100H, find the content of DL

```
X DW 567FH, 8AC0H,429CH,0017H
MOV BP,OFFSET X
MOV SI,3
MOV DL,[BP][SI]
```

A. 8AH
B. C0H
C. 42H
D. 00H

12- The value of CX and AL registers after executing the following code is:

```
MOV AL, FFH
MOV CX, 6
L1: ADD AL, 1
LOOPNZ L1
```

A. CX=4; AL=0
B. CX=5;AL=0;
C. CX=2;AL=1
D. CX=3; AL=1

13- LOPNE instruction repeat a group of instruction while

A. (CX=0)&(ZF=0)
B. (CX=0) & (ZF≠0)
C. (CX≠0) &(ZF=1)
D. (Cx≠0) & (ZF=0)

14- Find the contents of memory location X after execution of the following code:

```
X DB 'KLM+ABC'
CLD
MOV AL, 'Y'
MOV DI, OFFSET X
MOV CX, 3
L: STOSB
LOOP L
```

A. X='YYYYABC'
B. X=' KLM+YYY'
C. X='YYY+ABC'
D. X=' KLM+ABC'

15- Find the contents of registers AL and DL after execution of the following code:

```
X DB 'KLM+ABC'
Y DB 'KLM*WBD'
CLD
MOV SI, OFFSET X
MOV DI, OFFSET Y
MOV CX, 7
REPE CMPSB
MOV AL,[SI]
MOV DL,[DI]
```

A. AL='+'; DL='*'
B. AL='+'; DL='W'
C. AL='A';DL='*'
D. AL='A'; DL='W'

16- The last instruction in interrupt service routine (ISR) should be:

A. END
B. ENDS
C. IRET
D. INTR

17- Find the value of AL and AH registers after execution of the following assembler code:

```
MOV  AX,11
MOV  BL,3
CALL  XXX
MOV  BL,AL
MOV  DL, AH
HLT
XXX PROC
DIV  BL
RET
XXX ENDP
```

A. AL=3 AH=2   B.AL=2, AH=3   C. AL=4 ,AH=2   D. AL=2, AH=4

18- What is the value of SI after executing the following code if the DF=1?

```
MOV SI, 301H
LODSB
```

A. 300H   B. 2FFH   C. 301H   D. 299H

19- What is the value of CX REGISTER after execution the following codes?

```
STR DB "ABCDE"

MOV DI, OFFSET STR
MOV CX,5
MOV AL,'C'
REPNE SCASB
```

A.3   B. 2   C. 1   D.0

20- What will be appears on the screen after executing the following code:

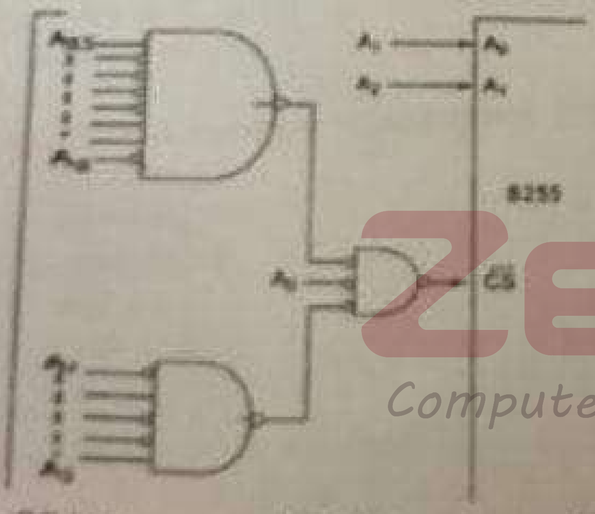
```
STR DB "CDEFGH", 0DH, "AB"
DB 'XYZ' , '$'
MOV AH, 9
MOV DX, OFFSET STR
INT 21H
```

A. CDEFGHAB   B. ABXYZH   C. ABCDEFGH   D.ABXYZ

21- During the execution near call instruction the ------------------changed

A. only IP   B. only CS   C. IP,CS   D. IP, DS

22- Use the following circuit of interfacing an 8255 with 8086. what is the address of port B



A. 864AH   B. 8649H   C.7982H   D.7981H

Question #2

- Write instructions to test bit-4 in register AL, if it is one reset bit-1 otherwise set bit-2

- Write a procedure to store '*' into 1 KB memory locations defined as byte;

```
X        DB 1024 DUP(?)
```

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- Write an assembler codes to move cursor to position (12,30) using BIOS interrupt

- Implement the following C++ statement using Intel 8086 assembler codes:
If (x>8) s=s+x; else s=s-x; where s, x memory locations defined as signed numbers

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- List 3 differences between Synchronous and Asynchronous serial communication.

QUESTION #1 (ANSWERS)

1	2	3	4	5	6	7	8	9	10	11
12	13	14	15	16	17	18	19	20	21	22

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Question #3

Interface an input port 74LS245 to read the status of switches SW1 to SW8. The switches, when shorted, input a 1 else input a 0 to the microprocessor system. Store the status into register DL, assuming the address of the port is 0840H.

A. Draw schematic diagram

B. Write assembler code to read status of switches and store the result into DL register